Neighborhood averaging accounts for spatial gains of a graph encoder in tumor spatial omics

Zeyu Fu

Supplementary methods

Display definitions. Maps show the predetermined display field (the largest retained field). Only reference classes with at least 1% prevalence over all valid observations of an object are eligible for cluster-color matching. With $n_{ab} = \sum_{i \in \mathcal{V}} \mathbf{1}(y_i = a, \widehat{y}_i = b)$ and $\mathcal{A}_{C_{\text{common}}}$ the one-to-one partial matchings restricted to those classes, the rectangular Hungarian assignment selects

$$C_{\text{common}} = \{a : n_{a\cdot}/|\mathcal{V}| \geq 0.01\}, \qquad \mathcal{M}^* = \arg \max_{\mathcal{M} \in \mathcal{A}_{C_{\text{common}}}} \sum_{(a,b) \in \mathcal{M}} n_{ab}. \tag{S1}$$

Rare reference classes, unmatched clusters and unknown reference labels are gray; all scores retain the original predictions and all valid classes. Class prevalence and matching use the complete object, not the display field. The colorectal overview keeps supplied broad compartments and maps eligible matched `1v2` labels to their modal supplied compartment. Category colors keep their original class index even when rare classes are removed from the legend.

Breast registration and region of interest. Spot centers are scaled to the high-resolution image and checked against the source tissue-position table. The fixed region of interest is centered at the median spot position:

$$\begin{aligned} r_i^{\text{hires}} &= s_{\text{hires}} r_i, \qquad \ell_{\text{pixels}}(L) = L\, s_{\text{hires}}/u_{\text{pixel}}, \\ \mathcal{R} &= \prod_{d=1}^{2} [m_d - 0.22 a_d,\, m_d + 0.22 a_d], \quad m_d = \text{median}_i\, r_{id}^{\text{hires}}, \quad a_d = \max_i r_{id}^{\text{hires}} - \min_i r_{id}^{\text{hires}}, \end{aligned} \tag{S2}$$

where L is a physical length in micrometers and u_{pixel} the source calibration. Colorectal local-millimeter coordinates are multiplied by 1,000 for the 100-micrometer bars; no physical scale is assigned to lung, BCC, Open-ST or MIBI maps.

Breast composition descriptors. With supplied composition fractions p_{ic} over C categories and the tumor set $\mathcal{T}$ (invasive, proliferating invasive and the two DCIS categories),

$$\begin{aligned} \eta_i &= \max_c p_{ic}, \qquad h_i = -\frac{\sum_c p_{ic} \log(p_{ic} + 10^{-12})}{\log C}, \qquad v_i = \sum_{c \in \mathcal{T}} p_{ic}, \\ M_{a,s}(\tau) &= \text{ARI}(y_{I_\tau}, \widehat{y}_{a,s,I_\tau}), \qquad I_\tau = \{i : \eta_i \geq \tau\}. \end{aligned} \tag{S3}$$

The seam-score association is the Spearman correlation of h_i and the node seam score b_i with average ranks for ties.

Histograms, color scales and execution records. Gate densities use 30 equal-width bins on $[0, 1]$, normalized separately per group, $f_k = N_k/(|A|w_k)$. For a measured channel v_i in display field D, colors use

$$w_i = \begin{cases} \log[1 + \max(v_i, 0)], & \text{protein intensity,} \\ v_i, & \text{saved RNA marker values,} \end{cases} \qquad c_i = \text{clip}_{[0,1]} \frac{w_i - Q_\alpha(w_D)}{Q_{0.98}(w_D) - Q_\alpha(w_D)}, \tag{S4}$$

with $\alpha = 0.02$ for proteins and $\alpha = 0$ for RNA. Run time is the recorded per-arm timer (fitting, K-means, export and scoring), and memory is the peak allocated GPU memory over seeds.

Scope. All 351 primary method–seed records and 45 transfer records were fitted once with the released hyperparameters and are unchanged. The follow-up analyses (tuned controls, re-scoring, corrected nulls, whole sections and transfer controls) import the frozen estimator code without modifying it. Three optimization seeds are not independent specimens; the five lung and four BCC patients support patient-level summaries in the primary panel. Whole-section controls and the 200-neighbor transfer control were informed by bounded analyses of the same patients; they are sensitivity or descriptive comparisons, not independently selected validation. The accompanying source package documents the numerical provenance and the distinction between document reconstruction and analysis reproduction.

Independent cohort and further tissues. The all-label comparison in the independent cohort was fixed in a written protocol before any fit. The boundary targets, weighting controls, smoothed seam scores and tissue panel followed the first results and are exploratory. The frozen estimator, graph rule and preprocessing were imported unchanged, every fit ran on CPU, and three repetitions of the unchanged protocol were retained because optimization is not bit-reproducible. Each repetition averages three seeds and fits one encoder per patient, distinct target class count and seed; targets with the same class count share that fit. Repetitions add no patients, and the same seeded control procedure is used in every repetition. Section identity for the one patient with differing image and label names was checked by array-coordinate joins to both public sections.

Complete primary partition scores

Table S1: ARI means ± seed SD: expression-only, simple spatial and gated arms.

Object	PCA50	PCA30	Neighbor mean	Tessera	No gate
Lung5_Rep2	0.059 ± 0.000	0.060 ± 0.001	0.208 ± 0.000	0.231 ± 0.039	0.288 ± 0.006
Lung6	0.377 ± 0.000	0.377 ± 0.000	0.417 ± 0.001	0.322 ± 0.004	0.322 ± 0.004
Lung9_Rep1	0.126 ± 0.000	0.125 ± 0.001	0.185 ± 0.000	0.198 ± 0.015	0.206 ± 0.004
Lung12	0.049 ± 0.000	0.048 ± 0.000	0.181 ± 0.000	0.190 ± 0.009	0.208 ± 0.002
Lung13	0.039 ± 0.000	0.039 ± 0.000	0.132 ± 0.000	0.191 ± 0.001	0.205 ± 0.008
BCC A	0.290 ± 0.013	0.308 ± 0.010	0.244 ± 0.025	0.289 ± 0.014	0.273 ± 0.005
BCC B	0.097 ± 0.003	0.097 ± 0.003	0.077 ± 0.001	0.073 ± 0.008	0.070 ± 0.001
BCC C	0.128 ± 0.008	0.122 ± 0.000	0.123 ± 0.000	0.099 ± 0.001	0.097 ± 0.001
BCC D	0.156 ± 0.000	0.156 ± 0.001	0.152 ± 0.033	0.105 ± 0.022	0.114 ± 0.011
Breast Visium	0.221 ± 0.011	0.212 ± 0.003	0.198 ± 0.015	0.154 ± 0.010	0.153 ± 0.011
TVA 232	0.094 ± 0.005	0.100 ± 0.002	0.114 ± 0.009	0.149 ± 0.007	0.149 ± 0.006
MIBI	0.210 ± 0.001	0.211 ± 0.001	0.149 ± 0.001	0.185 ± 0.042	0.180 ± 0.035
Open-ST HNSCC	0.271 ± 0.029	0.266 ± 0.003	0.211 ± 0.010	0.219 ± 0.022	0.236 ± 0.002

Table S2: ARI means ± seed SD: remaining component and geometry controls.

Object	No fusion	No contrast	Self-only	Permuted XY
Lung5_Rep2	0.283 ± 0.013	0.060 ± 0.002	0.062 ± 0.005	0.139 ± 0.005
Lung6	0.325 ± 0.003	0.373 ± 0.004	0.373 ± 0.005	0.370 ± 0.036
Lung9_Rep1	0.200 ± 0.004	0.133 ± 0.005	0.131 ± 0.003	0.022 ± 0.026
Lung12	0.204 ± 0.011	0.060 ± 0.008	0.063 ± 0.012	0.101 ± 0.012
Lung13	0.203 ± 0.005	0.052 ± 0.003	0.052 ± 0.001	0.004 ± 0.002
BCC A	0.287 ± 0.012	0.290 ± 0.014	0.291 ± 0.012	0.240 ± 0.005
BCC B	0.073 ± 0.008	0.076 ± 0.002	0.075 ± 0.001	0.055 ± 0.003
BCC C	0.096 ± 0.004	0.110 ± 0.014	0.110 ± 0.010	0.067 ± 0.010
BCC D	0.119 ± 0.001	0.148 ± 0.003	0.149 ± 0.003	0.094 ± 0.017
Breast Visium	0.161 ± 0.003	0.202 ± 0.005	0.198 ± 0.009	0.025 ± 0.012
TVA 232	0.141 ± 0.009	0.103 ± 0.011	0.101 ± 0.009	0.099 ± 0.017
MIBI	0.177 ± 0.042	0.209 ± 0.009	0.208 ± 0.006	0.056 ± 0.010
Open-ST HNSCC	0.217 ± 0.026	0.233 ± 0.002	0.236 ± 0.006	0.052 ± 0.030

Table S3: NMI means ± seed SD: expression-only, simple spatial and gated arms.

Object	PCA50	PCA30	Neighbor mean	Tessera	No gate
Lung5_Rep2	0.120 ± 0.000	0.119 ± 0.002	0.283 ± 0.000	0.334 ± 0.022	0.403 ± 0.013
Lung6	0.454 ± 0.000	0.454 ± 0.000	0.530 ± 0.000	0.468 ± 0.010	0.470 ± 0.005
Lung9_Rep1	0.170 ± 0.000	0.170 ± 0.001	0.282 ± 0.000	0.276 ± 0.023	0.291 ± 0.015
Lung12	0.129 ± 0.000	0.127 ± 0.000	0.285 ± 0.000	0.301 ± 0.017	0.333 ± 0.007
Lung13	0.066 ± 0.000	0.065 ± 0.000	0.162 ± 0.000	0.232 ± 0.003	0.248 ± 0.011
BCC A	0.499 ± 0.004	0.510 ± 0.005	0.391 ± 0.007	0.491 ± 0.004	0.476 ± 0.008
BCC B	0.341 ± 0.003	0.341 ± 0.004	0.287 ± 0.001	0.309 ± 0.012	0.302 ± 0.005
BCC C	0.375 ± 0.010	0.368 ± 0.001	0.338 ± 0.000	0.363 ± 0.003	0.365 ± 0.005
BCC D	0.364 ± 0.009	0.369 ± 0.001	0.338 ± 0.013	0.347 ± 0.019	0.361 ± 0.007
Breast Visium	0.407 ± 0.012	0.396 ± 0.001	0.352 ± 0.002	0.357 ± 0.010	0.361 ± 0.008
TVA 232	0.346 ± 0.007	0.347 ± 0.005	0.322 ± 0.003	0.369 ± 0.005	0.368 ± 0.007
MIBI	0.324 ± 0.002	0.325 ± 0.001	0.264 ± 0.001	0.262 ± 0.019	0.261 ± 0.016
Open-ST HNSCC	0.428 ± 0.001	0.423 ± 0.006	0.372 ± 0.001	0.394 ± 0.011	0.402 ± 0.005

Table S4: NMI means ± seed SD: remaining component and geometry controls.

Object	No fusion	No contrast	Self-only	Permuted XY
Lung5_Rep2	0.390 ± 0.016	0.120 ± 0.002	0.124 ± 0.004	0.222 ± 0.010
Lung6	0.469 ± 0.009	0.453 ± 0.003	0.453 ± 0.003	0.506 ± 0.012
Lung9_Rep1	0.288 ± 0.004	0.173 ± 0.003	0.172 ± 0.003	0.030 ± 0.037
Lung12	0.323 ± 0.017	0.138 ± 0.011	0.137 ± 0.006	0.172 ± 0.005
Lung13	0.249 ± 0.011	0.074 ± 0.003	0.075 ± 0.004	0.008 ± 0.003
BCC A	0.493 ± 0.008	0.499 ± 0.007	0.504 ± 0.010	0.414 ± 0.008
BCC B	0.306 ± 0.010	0.312 ± 0.006	0.310 ± 0.002	0.236 ± 0.016
BCC C	0.360 ± 0.006	0.354 ± 0.018	0.361 ± 0.011	0.235 ± 0.027
BCC D	0.353 ± 0.009	0.354 ± 0.000	0.352 ± 0.015	0.282 ± 0.018
Breast Visium	0.361 ± 0.006	0.404 ± 0.011	0.405 ± 0.009	0.093 ± 0.024
TVA 232	0.364 ± 0.001	0.322 ± 0.006	0.316 ± 0.013	0.288 ± 0.018
MIBI	0.256 ± 0.028	0.324 ± 0.004	0.328 ± 0.004	0.086 ± 0.013
Open-ST HNSCC	0.393 ± 0.014	0.400 ± 0.004	0.398 ± 0.004	0.095 ± 0.039

Table S5: Patient-held-out transfer: accuracy / balanced accuracy, three-seed means for each held-out patient.

Target patient	Target	PCA50	Self-only	Tessera
Lung5_Rep2	Region	0.227 / 0.235	0.218 / 0.230	0.381 / 0.314
Lung6	Region	0.544 / 0.292	0.661 / 0.312	0.761 / 0.392
Lung9_Rep1	Region	0.226 / 0.321	0.268 / 0.328	0.181 / 0.342
Lung12	Region	0.212 / 0.225	0.193 / 0.217	0.180 / 0.278
Lung13	Region	0.304 / 0.210	0.341 / 0.201	0.408 / 0.233
Lung5_Rep2	Cell type	0.559 / 0.398	0.506 / 0.357	0.396 / 0.253
Lung6	Cell type	0.377 / 0.412	0.358 / 0.434	0.324 / 0.311
Lung9_Rep1	Cell type	0.258 / 0.227	0.261 / 0.226	0.248 / 0.157
Lung12	Cell type	0.492 / 0.278	0.428 / 0.243	0.356 / 0.167
Lung13	Cell type	0.399 / 0.341	0.445 / 0.368	0.372 / 0.204

Table S6: Last-layer gate diagnostics, mean over three seeds. AUROCs measure discrimination of edges whose endpoints carry different supplied labels, with ties counted as one half. "Expression difference" scores each edge by the Euclidean distance between its endpoints' PCA features.

Object	Gate AUROC	Expression-difference AUROC	Median gate	Gates < 0.05 (%)
Lung5_Rep2	0.496	0.473	0.011	81
Lung6	0.385	0.623	0.047	58
Lung9_Rep1	0.475	0.500	0.091	50
Lung12	0.495	0.536	0.015	77
Lung13	0.548	0.559	0.052	58
BCC A	0.621	0.642	0.060	54
BCC B	0.485	0.527	0.032	67
BCC C	0.511	0.527	0.112	40
BCC D	0.498	0.528	0.069	40
Breast Visium	0.467	0.662	0.244	15
Colorectal TVA	0.444	0.450	0.186	32
MIBI	0.420	0.726	0.199	23
Open-ST HNSCC	0.546	0.680	0.040	65

Cluster-number and relative-weight sensitivity

Table S7: Saved embeddings re-clustered with K_{common}, the number of supplied classes with at least 1% object-level prevalence. ARI retains all valid labels, including rare classes. Values are three-seed means; Δ is Tessera minus PCA50. This does not tune neighborhood scales or change the primary predictions.

Object	Primary K	K_{common}	PCA50	Mean $k = 6$	Tessera	Δ
Lung5_Rep2	7	6	0.032	0.232	0.268	+0.236
Lung6	5	4	0.409	0.455	0.414	+0.005
Lung9_Rep1	5	4	0.116	0.209	0.190	+0.074
Lung12	7	7	0.049	0.181	0.190	+0.141
Lung13	5	3	0.016	0.016	0.388	+0.372
BCC A	12	8	0.308	0.274	0.343	+0.035
BCC B	11	5	0.192	0.136	0.153	-0.038
BCC C	12	6	0.216	0.309	0.246	+0.030
BCC D	11	9	0.218	0.239	0.142	-0.076
Breast Visium	14	9	0.278	0.247	0.197	-0.081
Colorectal TVA	34	21	0.126	0.133	0.184	+0.058
MIBI	8	8	0.210	0.149	0.185	-0.025
Open-ST HNSCC	14	11	0.324	0.258	0.261	-0.063

Table S8: Relative last-layer gate weights over directed multiset edge entries. D is total-variation distance from uniform incident weights; R is effective neighbor count divided by multiset degree. Node medians and 90th percentiles are averaged over seeds; attenuation is the maximum $10^{-8}/(G_i + 10^{-8})$ over all nodes and seeds. All 39 saved object–seed records have zero isolated and zero zero-mass nodes. Small gates do not imply uniform weighting.

Object	Median D	90th percentile D	Median R	Max. attenuation
Lung5_Rep2	0.407	0.593	0.499	1.6e-05
Lung6	0.332	0.494	0.604	8.3e-06
Lung9_Rep1	0.318	0.470	0.629	2.8e-06
Lung12	0.382	0.559	0.533	1.4e-05
Lung13	0.323	0.492	0.621	4.3e-06
BCC A	0.271	0.430	0.707	4.9e-06
BCC B	0.307	0.465	0.647	1.7e-06
BCC C	0.260	0.390	0.719	3.4e-07
BCC D	0.275	0.419	0.693	4.3e-07
Breast Visium	0.187	0.295	0.827	5.9e-07
Colorectal TVA	0.200	0.327	0.806	4.7e-07
MIBI	0.244	0.373	0.739	5.9e-07
Open-ST HNSCC	0.329	0.486	0.609	1.5e-05

Independent cohort and further tissues

Table S9: Pathologist annotation of the eight independent sections. Matched is the percentage of labeled spots found in the section's own count matrix by array coordinate. Unassigned spots (undetermined) stay in the graph and the principal components; they are a class for the all-label target and missing for the other targets.

Section	Labeled rows	Matched (%)	Invasive	In situ	Connective	Adipose	Immune	Glands	Undetermined
A1	348	99.4	282	21	24	12	2	0	5
B1	295	100.0	63	0	95	70	0	41	26
C1	177	99.4	127	0	37	3	0	0	9
D1	309	99.0	139	0	109	0	7	0	51
E1	587	100.0	340	0	17	0	177	0	53
F1	692	100.0	577	0	56	0	26	0	32
G2	475	100.0	140	20	110	98	17	17	65
H1	613	100.0	90	97	169	112	23	39	83

Table S10: Paired contrasts over the eight patients for each target and for the weighting controls. Counts give patients above and below zero; intervals are 95% percentile bootstrap intervals over patients, conditional on the selected controls and without repeating configuration selection; p is the exact two-sided sign-flip value (smallest attainable 0.0078). Mean is the unweighted neighbor mean on Tessera's graph and weighted uses expression-similarity weights. The two variants restrict the tumor class to invasive cancer or drop breast glands.

Target	Contrast	Above/below	Mean	95% interval	p
All labels	Tessera − same-graph mean	1/7	-0.047	[-0.074, -0.020]	0.0156
	Tessera − tuned control	2/6	-0.022	[-0.047, +0.002]	0.141
	Same-graph mean − tuned control	6/2	+0.024	[+0.004, +0.047]	0.0781
	Weighted − same-graph mean	5/3	+0.003	[-0.005, +0.011]	0.438
	Weighted − shuffled weights	6/2	+0.005	[-0.006, +0.015]	0.469
	Weighted − shuffled coordinates	8/0	+0.194	[+0.120, +0.261]	0.00781
Named compartments	Tessera − same-graph mean	2/6	-0.028	[-0.075, +0.033]	0.367
	Tessera − tuned control	3/5	-0.020	[-0.067, +0.042]	0.562
	Same-graph mean − tuned control	6/2	+0.008	[-0.032, +0.043]	0.711
	Weighted − same-graph mean	6/2	+0.004	[-0.008, +0.014]	0.508
	Weighted − shuffled weights	6/2	+0.005	[-0.007, +0.016]	0.438
	Weighted − shuffled coordinates	8/0	+0.216	[+0.125, +0.304]	0.00781
Tumor boundary	Tessera − same-graph mean	3/5	-0.062	[-0.132, -0.000]	0.133
	Tessera − tuned control	4/4	-0.008	[-0.080, +0.059]	0.836
	Same-graph mean − tuned control	8/0	+0.053	[+0.023, +0.088]	0.00781
	Weighted − same-graph mean	4/4	-0.012	[-0.029, +0.005]	0.258
	Weighted − shuffled weights	4/4	-0.003	[-0.025, +0.016]	0.805
	Weighted − shuffled coordinates	8/0	+0.309	[+0.157, +0.456]	0.00781
Invasive cancer only	Tessera − same-graph mean	3/5	-0.073	[-0.171, +0.005]	0.172
	Tessera − tuned control	4/4	-0.015	[-0.094, +0.057]	0.719
	Same-graph mean − tuned control	8/0	+0.058	[+0.022, +0.100]	0.00781
	Weighted − same-graph mean	3/4	-0.014	[-0.030, +0.003]	0.188
	Weighted − shuffled weights	4/4	-0.001	[-0.024, +0.020]	0.914
	Weighted − shuffled coordinates	8/0	+0.327	[+0.166, +0.485]	0.00781
Without breast glands	Tessera − same-graph mean	3/5	-0.069	[-0.138, -0.005]	0.109
	Tessera − tuned control	3/5	-0.020	[-0.087, +0.044]	0.602
	Same-graph mean − tuned control	8/0	+0.049	[+0.022, +0.079]	0.00781
	Weighted − same-graph mean	4/4	-0.008	[-0.023, +0.006]	0.398
	Weighted − shuffled weights	6/2	+0.001	[-0.021, +0.020]	0.938
	Weighted − shuffled coordinates	8/0	+0.320	[+0.160, +0.479]	0.00781

Table S11: Edge-level seam AUROC per patient, mean over three seeds for the learned gate. A seam is a graph edge whose two valid endpoints carry different labels. Diff. is the plain feature difference; Smooth and Wtd. are the distance between neighborhood means of the two endpoints, unweighted or weighted by expression similarity.

	Tumor boundary				Named compartments			
Patient	Gate	Diff.	Smooth	Wtd.	Gate	Diff.	Smooth	Wtd.
A1	0.540	0.600	0.627	0.645	0.562	0.597	0.636	0.650
B1	0.458	0.412	0.510	0.507	0.553	0.600	0.610	0.617
C1	0.576	0.663	0.597	0.627	0.556	0.663	0.597	0.627
D1	0.490	0.525	0.579	0.586	0.539	0.565	0.601	0.603
E1	0.389	0.362	0.411	0.419	0.387	0.357	0.402	0.410
F1	0.505	0.464	0.476	0.484	0.503	0.453	0.469	0.475
G2	0.499	0.563	0.637	0.657	0.510	0.617	0.639	0.658
H1	0.580	0.575	0.706	0.740	0.562	0.551	0.570	0.599

Table S12: Three further tissues, three-seed mean ARI. Contig. is the fraction of valid six-neighbor edges whose endpoints share a label; Mean is the same-graph mean. The tuned control was selected on the other two tissues; Best 30 is the best of 30 configurations chosen with the test labels, a descriptive maximum within the tested grid. Gate and Smooth are seam AUROCs of the learned gate and of the smoothed feature difference. Each tissue was also used when the encoder was first benchmarked.

Tissue	Obs.	Classes	Contig.	PCA50	Mean	Tuned	Best 30	Tessera	Gate	Smooth
Human cortex (Visium)	3611	7	0.905	0.190	0.253	0.286	0.341	0.416	0.457	0.492
Hypothalamus (MERFISH)	5488	8	0.931	0.079	0.175	0.425	0.460	0.357	0.468	0.633
Mouse cortex (STARmap)	1207	7	0.921	0.163	0.491	0.568	0.606	0.505	0.480	0.585

Table S13: Three repetitions of the unchanged encoder fitting protocol, each averaging the same three seeds; the third supplies the reported values. Repetitions add no patients. Range is the largest minus the smallest replicate. The same-graph mean and the tuned control are deterministic.

Target	Patient	Run 1	Run 2	Run 3	Range	Same-graph mean	Tuned control
All labels	A1	0.104	0.104	0.101	0.003	0.187	0.183
	B1	0.268	0.267	0.267	0.001	0.279	0.281
	C1	0.141	0.139	0.139	0.002	0.251	0.180
	D1	0.181	0.179	0.179	0.003	0.254	0.221
	E1	0.018	0.022	0.022	0.004	0.017	0.008
	F1	0.047	0.047	0.047	0.000	0.055	0.053
	G2	0.168	0.168	0.171	0.003	0.215	0.219
	H1	0.299	0.287	0.305	0.018	0.344	0.264
Compartments	A1	0.148	0.145	0.146	0.002	0.233	0.232
	B1	0.244	0.244	0.244	0.000	0.366	0.331
	C1	0.145	0.145	0.152	0.007	0.220	0.145
	D1	0.187	0.188	0.188	0.001	0.219	0.257
	E1	0.172	0.165	0.169	0.007	0.016	0.005
	F1	0.050	0.049	0.049	0.001	0.028	0.134
	G2	0.210	0.213	0.210	0.003	0.273	0.212
	H1	0.399	0.400	0.398	0.002	0.422	0.398
Tumor boundary	A1	0.117	0.166	0.170	0.052	0.413	0.307
	B1	0.734	0.734	0.734	0.000	0.711	0.672
	C1	0.121	0.121	0.121	0.000	0.071	-0.007
	D1	0.217	0.220	0.217	0.003	0.383	0.377
	E1	0.010	0.010	0.011	0.001	0.045	0.034
	F1	0.009	0.009	0.009	0.000	0.093	0.087
	G2	0.417	0.414	0.410	0.007	0.385	0.349
	H1	0.579	0.578	0.579	0.002	0.645	0.501

Untouched colorectal pathology pilot

The protocol preceded download and label inspection. The explicit `exclude` category triggered the disclosed sensitivity after the first PCA/mean scores, before any seam AUROC. Tissue spots remain in the graph; `exclude` and missing labels are omitted only from sensitivity scores. Mixed tumor/stroma and immune-infiltration classes remain distinct. The three-patient percentile bootstrap is descriptive and unstable, and the minimum two-sided sign-flip value is 0.25. The fixed two-class auxiliary fitting head is not part of the training loss. ARI uses a target-class-informed cluster number after fitting. Seam AUROC uses unique undirected edges, so this pilot is not pooled with the earlier 24-object landscape.

Table S14: QC-filtered colorectal pilot, three-seed mean ARI and seam AUROC. Tissue is the retained graph size; Scored excludes the explicit QC category and missing labels. Mean is the untuned mean on Tessera's graph; Raw and Smooth are distances between PCA features and same-graph means. The section for A120838 is replicate 1; A595688 and A551763 use replicate 2. Every annotated tissue barcode matched exactly.

			ARI				Seam AUROC		
Patient	Tissue	Scored	PCA50	Mean	Tessera	NoGate	Gate	Raw	Smooth
A120838	328	240	0.271	0.223	0.206	0.211	0.507	0.499	0.528
A595688	387	356	0.481	0.338	0.301	0.303	0.648	0.646	0.589
A551763	1219	1162	0.074	0.025	0.001	0.000	0.546	0.575	0.547

Table S15: Original locked colorectal target, retaining the released `exclude` class. These scores preserve the original protocol but do not interpret QC seams as biological boundaries. The model fits are identical to those used for the sensitivity in Table S14; only the scored target and clustering resolution change.

Patient	Tessera ARI	Mean ARI	Gate AUROC	Raw AUROC
A120838	0.193	0.199	0.507	0.481
A595688	0.292	0.311	0.655	0.648
A551763	0.011	0.018	0.533	0.569

The 1,000 label-shuffle iterations per patient and target give null median AUROCs between 0.45 and 0.55 for every score. Their edge-level distributions are negative controls, not patient confidence intervals. Public Space Ranger H&E overlays were inspected for gross position, orientation and image bounds. This author check neither adjudicates compartment diagnoses nor establishes blinded inter-reader agreement. Complete tables, nulls, fit receipts and registration overlays are retained in the separate pilot directory.



Five NSCLC and four BCC patients; other objects do not add patient-level replication

Figure S1: Scope of the bounded inputs. (a) Source and modeled observations on a logarithmic axis. (b) Retained coordinate frames. (c) Observed reference classes. (d) Fraction of total variance of the standardized, clipped input retained by the common 50-component PCA.

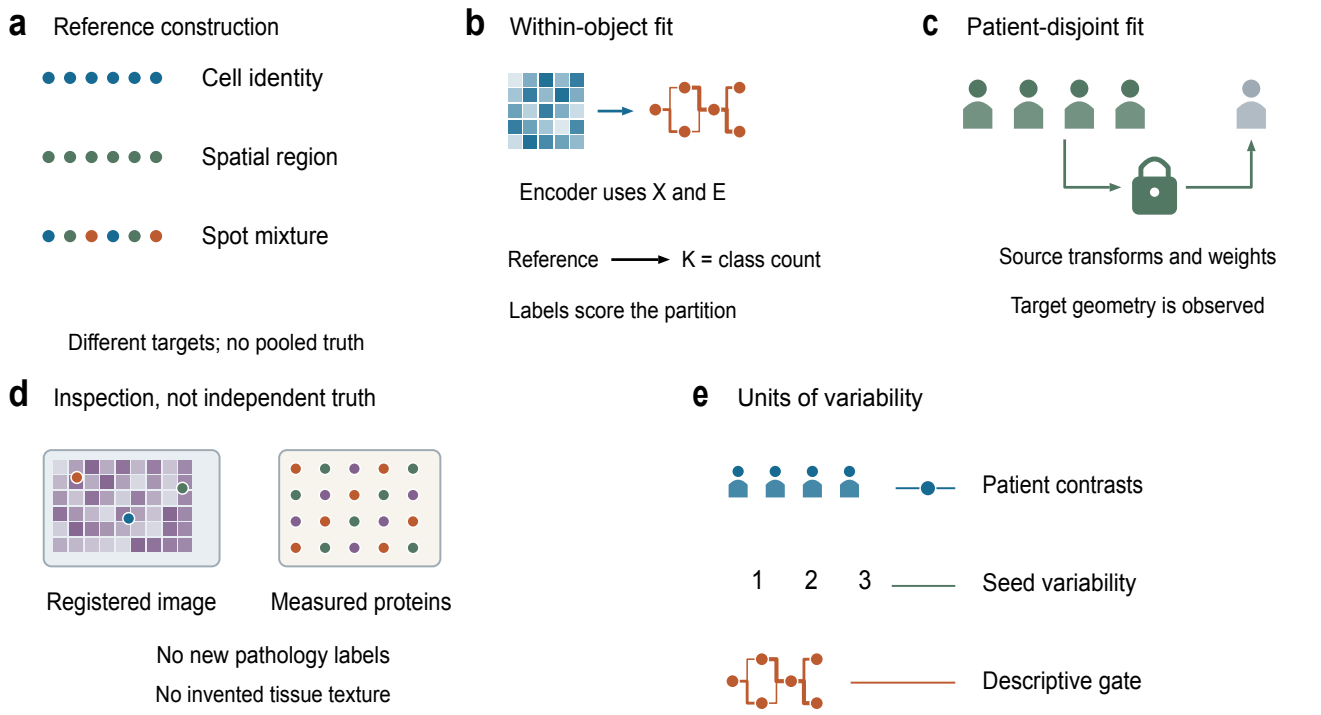


Figure S2: Targets and information access, conceptual. (a) Molecular cell categories, neighborhood-derived regions and mixed-spot compositions come from different constructions. (b) Within-object clustering uses the reference class count but no reference labels. (c) Patient transfer freezes all fitted quantities before the held-out patient is mapped. (d) Registered images and protein channels support inspection only. (e) Patient contrasts, seed variation and gate readouts answer different questions.

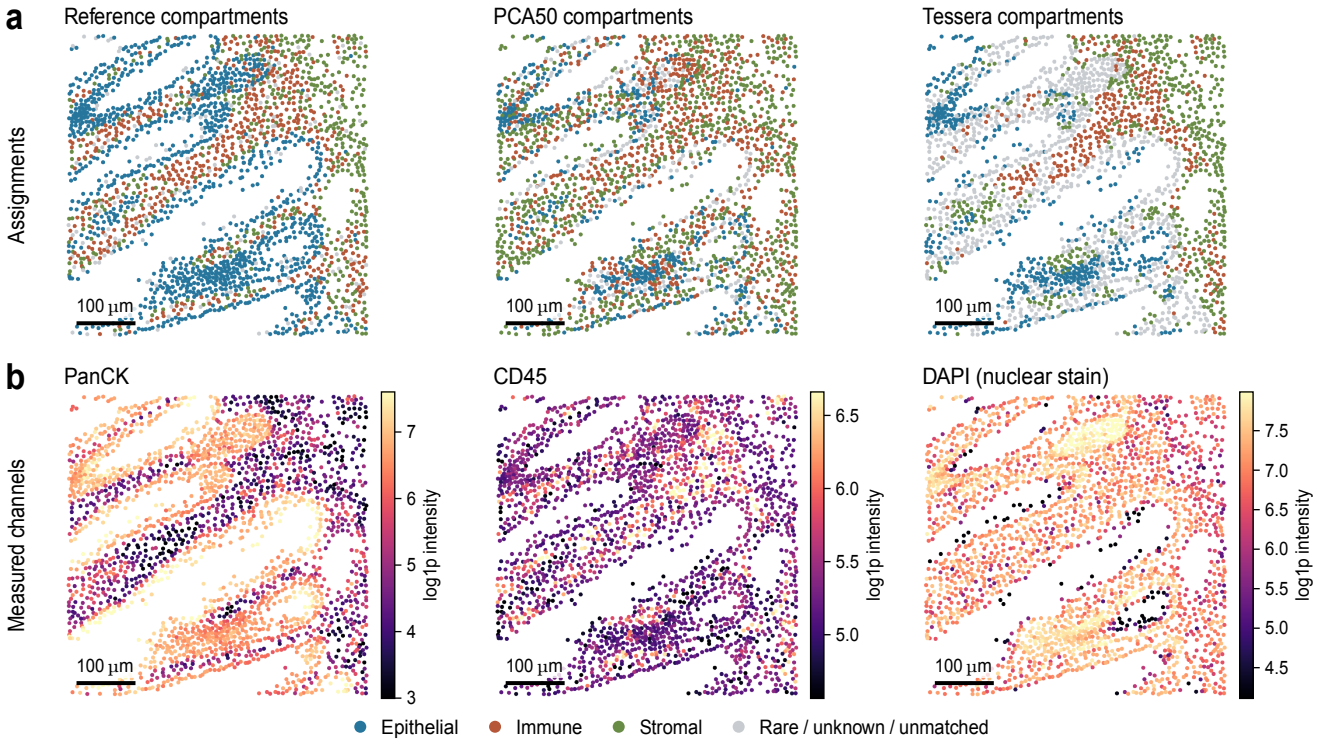


Figure S3: Colorectal TVA section 232, display field. (a) Supplied compartments and display-matched PCA50 and Tessera assignments; only `lv2` classes with at least 1% object prevalence are eligible for matching before modal compartment mapping, and unmatched clusters are gray. (b) Measured PanCK, CD45 and DAPI mean intensities, shown as $\log(1+x)$ with field-wise 2nd–98th percentile color limits.

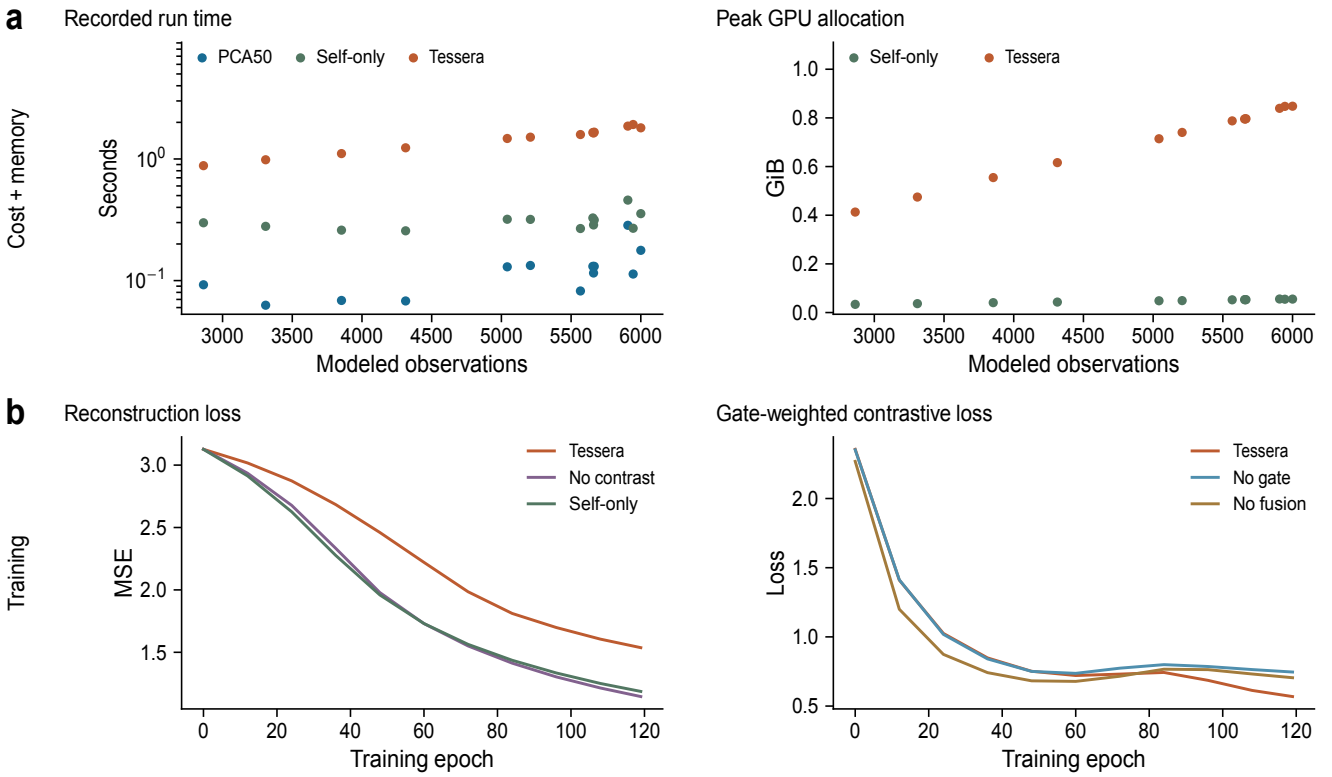


Figure S4: Execution and training records. (a) Recorded run time against modeled observations and peak GPU allocation. (b) Reconstruction and gate-weighted contrastive losses for Lung5_Rep2, seed 1, at the saved epochs.

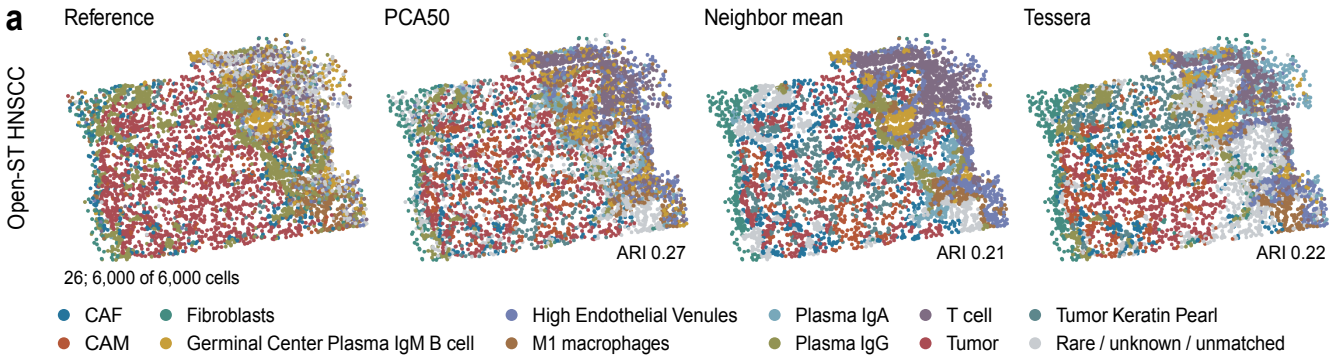


Figure S5: Open-ST section 26, complete selected 6,000-observation subset. (a) Supplied annotations, PCA50, the six-neighbor mean and Tessera on identical coordinates (seed 1). Matching excludes classes below 1% object prevalence; rare reference labels, unknowns and unmatched clusters are gray. ARI badges retain original predictions and all valid classes, averaged over three seeds.



Figure S6: Three MIBI fields in rows (a–c). Columns show supplied molecular categories, PCA50, the six-neighbor mean and Tessera (seed 1). Matching excludes reference classes below 1% object prevalence; rare reference labels, unknowns and unmatched clusters are gray. Fields are not independent patients.

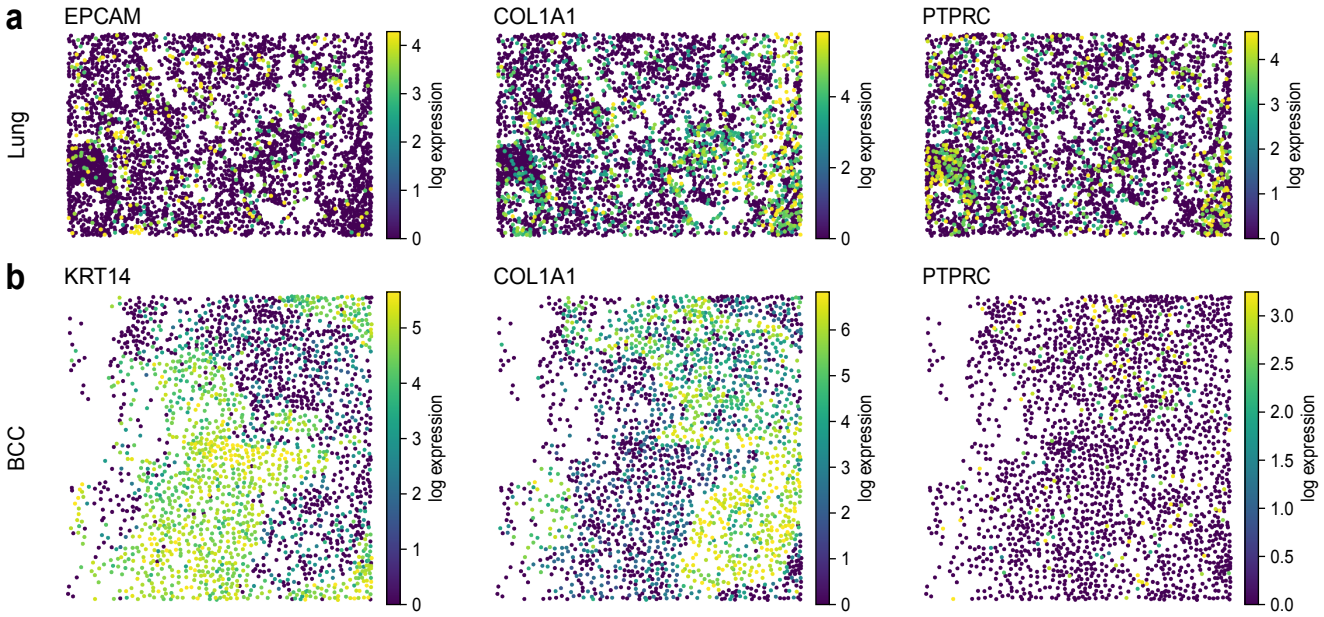


Figure S7: Measured RNA markers in the display fields. (a) Lung EPCAM, COL1A1 and PTPRC. (b) BCC KRT14, COL1A1 and PTPRC. Values are the saved log-normalized expression with 0–98th percentile color limits; high-expressing cells are drawn on top.

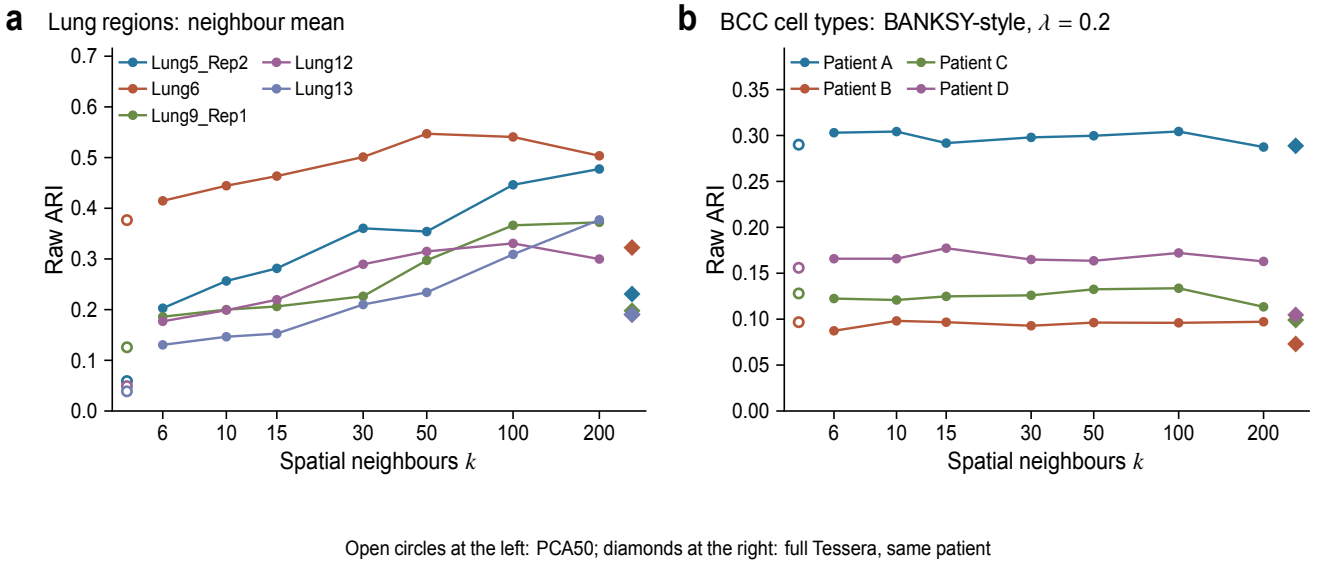


Figure S8: Agreement of simple spatial controls as a function of neighborhood size. (a) Lung region ARI of the neighbor mean. (b) BCC cell-type ARI of the BANKSY-style representation with $\lambda = 0.2$. Each line is one patient (three-seed mean); open circles at the left mark PCA50 and diamonds at the right mark full Tessera for the same patient.

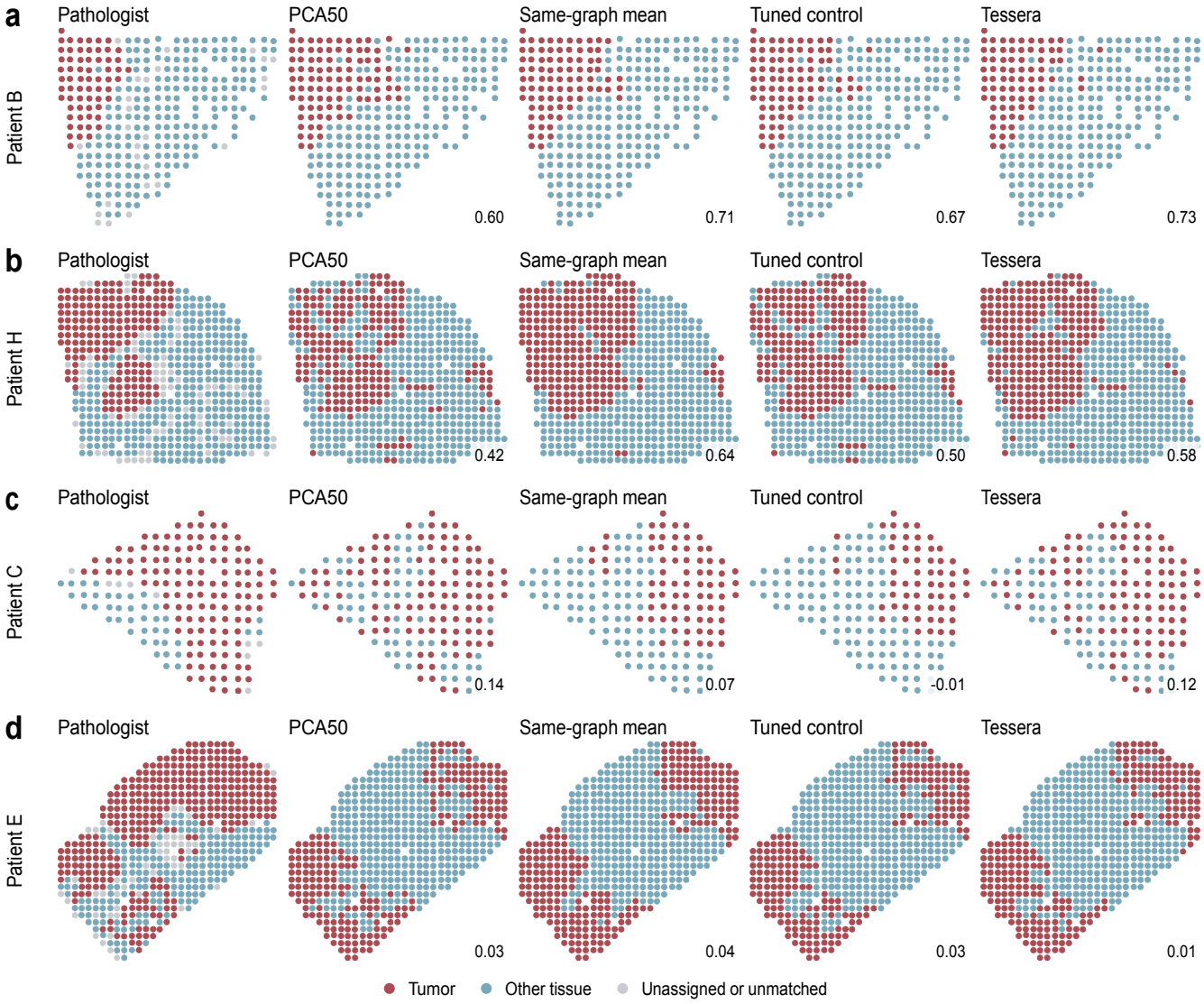


Figure S9: Tumor boundary in four of the eight independent patients: rows (a–d) are the two patients with the highest (B, H) and the two with the lowest (C, E) same-graph mean agreement. Columns show the pathologist's regions, PCA50, the unweighted mean of neighbors on Tessera's graph, a tuned control selected on the other seven patients, and Tessera (seed 1). Clusters are matched to the two classes for display only; gray marks spots the pathologist left unassigned and unmatched clusters. Badges give the three-seed mean ARI against tumor versus other tissue.

Additional supporting displays

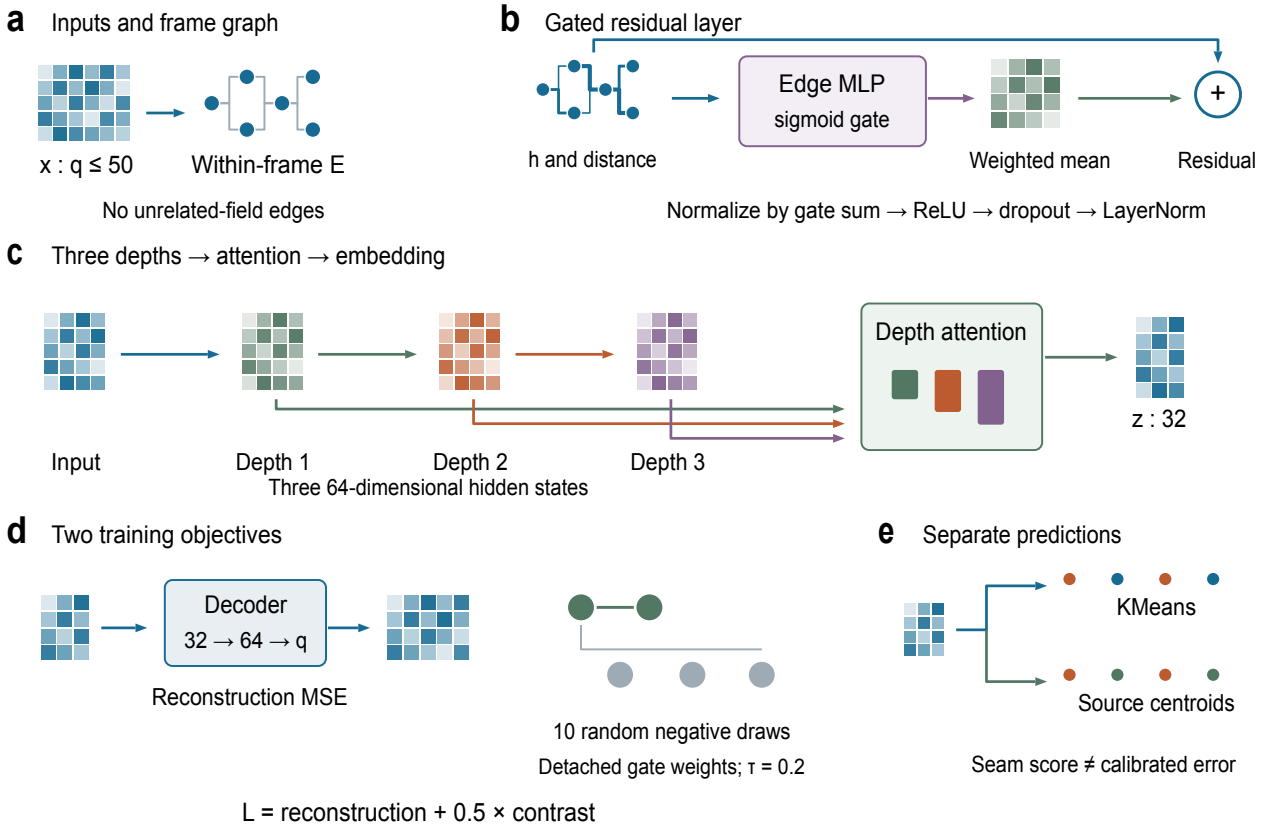


Figure S10: The Tessera computation as released. (a) Common PCA features and frame-restricted neighborhoods. (b) Edge differences and distances determine gates; gated messages are normalized before a residual update. (c) Three depths feed attention and a 32-dimensional projection. (d) A decoder reconstructs the PCA features; neighbor-positive contrast is weighted by detached gates. (e) K-means returns the within-object partition, and source-category centroids return transfer labels. Glyphs are conceptual; dimensions are those implemented.

Table S16: Three-seed mean ARI on the 13 objects with the primary cluster number. Δ is Tessera minus PCA50.

Object	PCA50	Neighbor mean	Tessera	Self-only	Δ
Lung5_Rep2	0.059	0.208	0.231	0.062	+0.172
Lung6	0.377	0.417	0.322	0.373	-0.054
Lung9_Rep1	0.126	0.185	0.198	0.131	+0.072
Lung12	0.049	0.181	0.190	0.063	+0.141
Lung13	0.039	0.132	0.191	0.052	+0.152
BCC A	0.290	0.244	0.289	0.291	-0.001
BCC B	0.097	0.077	0.073	0.075	-0.024
BCC C	0.128	0.123	0.099	0.110	-0.029
BCC D	0.156	0.152	0.105	0.149	-0.051
Breast Visium	0.221	0.198	0.154	0.198	-0.067
TVA 232	0.094	0.114	0.149	0.101	+0.056
MIBI	0.210	0.149	0.185	0.208	-0.025
Open-ST HNSCC	0.271	0.211	0.219	0.236	-0.052

Table S17: Tuned spatial controls and whole sections. Bounded-field controls maximize mean ARI over the other patients; values are three-seed means. Whole sections use all eligible cells and fixed cohort-modal controls (200-neighbor mean for NSCLC; BANKSY-style, $k = 100$, $\lambda = 0.2$ for BCC), informed by the bounded analysis of the same patients. Whole-section values are sensitivity results, not independently nested selections.

		Bounded fields			Whole sections			
Patient	Selected control	PCA50	Control	Tessera	Cells	PCA50	Control	Tessera
Lung5_Rep2	neighbour mean, $k = 200$	0.059	0.477	0.231	100,335	0.121	0.537	0.269
Lung6	neighbour mean, $k = 200$	0.377	0.504	0.322	89,975	0.131	0.206	0.054
Lung9_Rep1	neighbour mean, $k = 200$	0.126	0.372	0.198	87,606	0.099	0.271	0.179
Lung12	neighbour mean, $k = 200$	0.049	0.300	0.190	71,304	0.037	0.227	0.125
Lung13	neighbour mean, $k = 100$	0.039	0.309	0.191	81,236	0.038	0.212	0.090
BCC A	BANKSY-style, $k = 100$, $\lambda = 0.2$	0.290	0.304	0.289	21,368	0.240	0.235	0.227
BCC B	BANKSY-style, $k = 100$, $\lambda = 0.2$	0.097	0.096	0.073	41,758	0.073	0.071	0.054
BCC C	BANKSY-style, $k = 100$, $\lambda = 0.2$	0.128	0.134	0.099	90,636	0.092	0.105	0.060
BCC D	BANKSY-style, $k = 100$, $\lambda = 0.2$	0.156	0.172	0.105	65,152	0.119	0.103	0.082

Table S18: Independent pathologist cohort, three-seed mean ARI. The tuned control for each patient was selected on the other seven patients; the same-graph mean averages neighbors on Tessera's graph without tuning. Section identifiers are in parentheses; Classes counts the annotation classes scored for all labels.

			All labels				Tumor boundary		
Patient	Spots	Classes	PCA50	Same-graph mean	Tuned control	Tessera	Same-graph mean	Tuned control	Tessera
A (A1)	346	6	0.216	0.187	0.183	0.101	0.413	0.307	0.170
B (B1)	295	5	0.245	0.279	0.281	0.267	0.711	0.672	0.734
C (C1)	176	4	0.145	0.251	0.180	0.139	0.071	-0.007	0.121
D (D1)	306	4	0.223	0.254	0.221	0.179	0.383	0.377	0.217
E (E1)	587	4	0.058	0.017	0.008	0.022	0.045	0.034	0.011
F (F1)	691	4	0.071	0.055	0.053	0.047	0.093	0.087	0.009
G (G2)	467	7	0.176	0.215	0.219	0.171	0.385	0.349	0.410
H (H1)	613	7	0.219	0.344	0.264	0.305	0.645	0.501	0.579

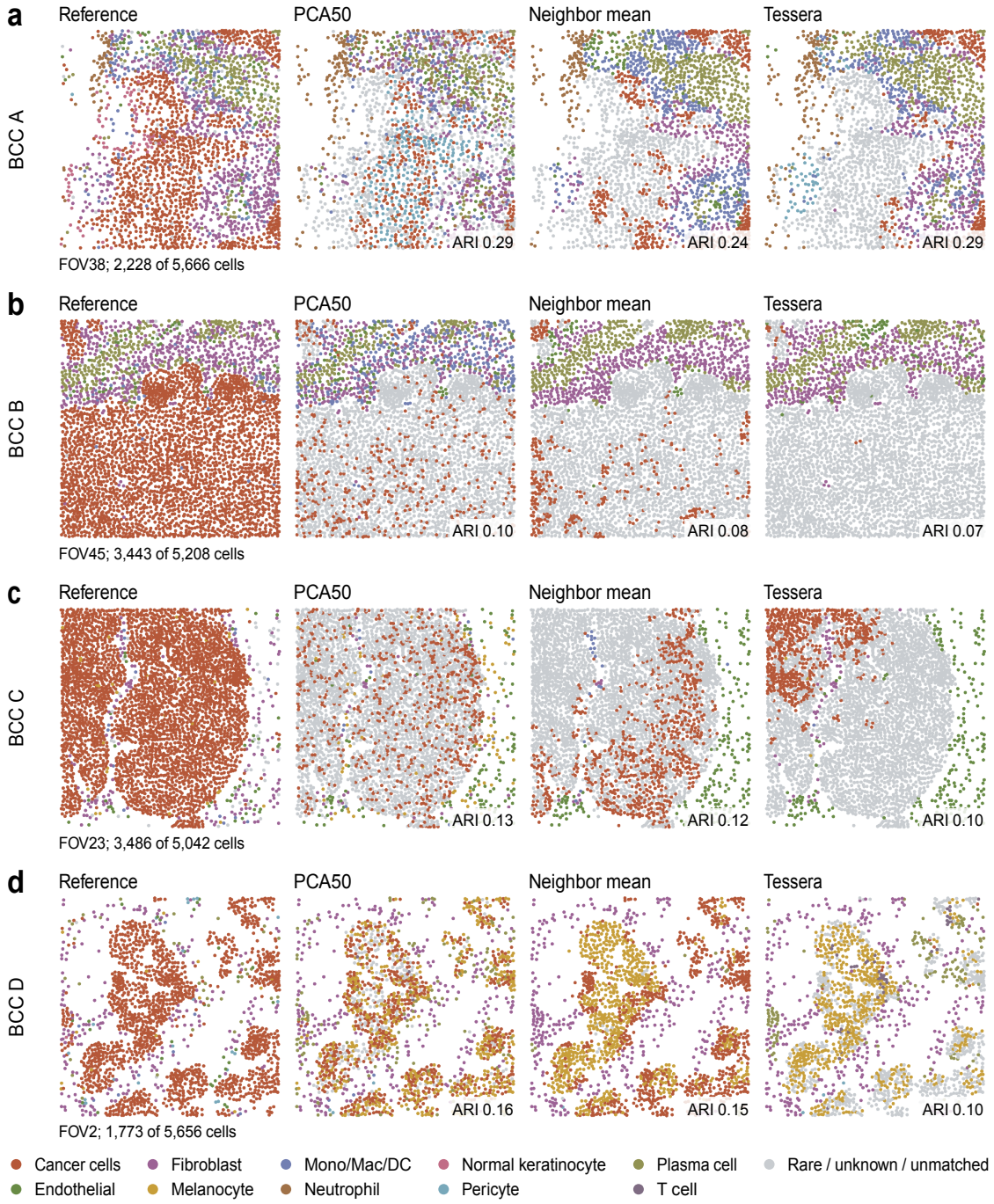


Figure S11: BCC display fields. Rows (a–d) show Patients A–D; columns show supplied cell types, PCA50, the six-neighbor mean and Tessera. Seed-1 clusters are matched only to reference classes with at least 1% object-level prevalence; rare reference classes, unknown labels and unmatched clusters are gray. ARI badges retain original predictions and all valid labels, averaged over three seeds and all retained fields.

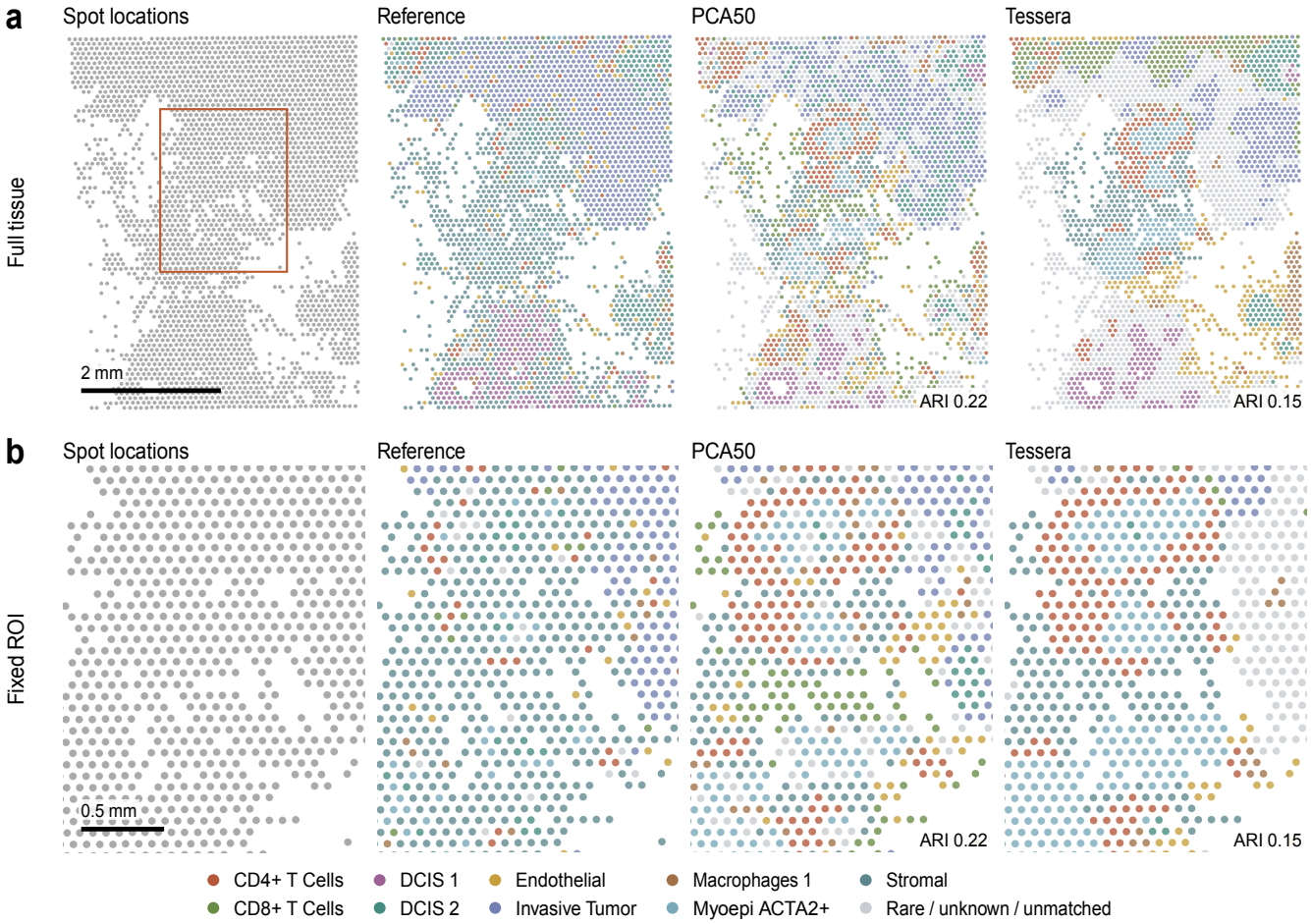


Figure S12: Breast spatial coordinates and matched representations. Row (a) shows full tissue and row (b) a median-centered ROI with fixed 22% half-extent, chosen without labels or scores. Columns show measured spot locations, the dominant Xenium-derived reference, PCA50 and Tessera (seed 1). All panels are author-generated coordinate maps; no source histology image is reproduced. Matching excludes reference categories below 1% object prevalence; rare reference labels, unknowns and unmatched clusters are gray. ARI badges retain all valid classes and are three-seed means. Scale bars use source calibration.

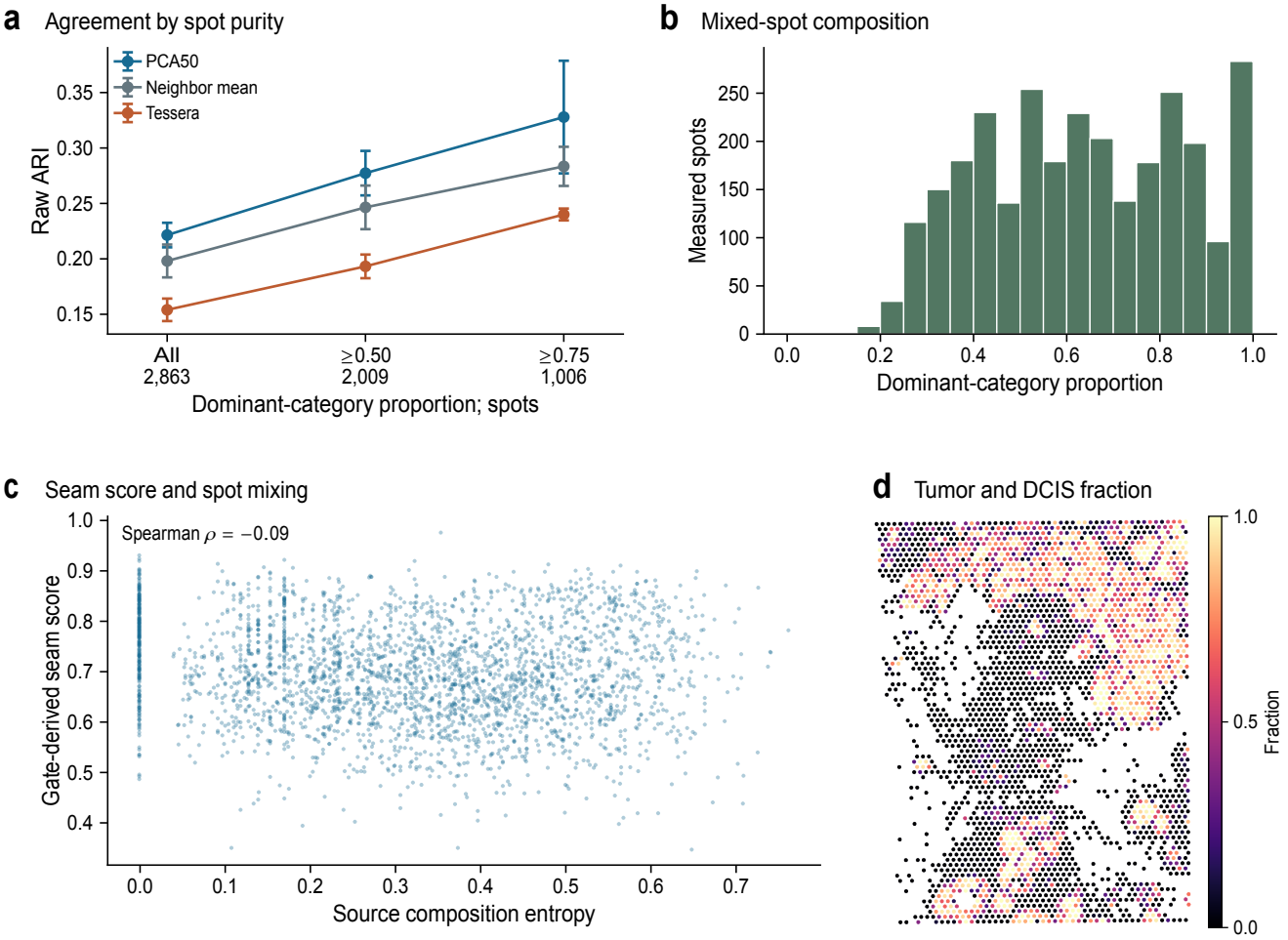


Figure S13: Breast mixed spots. (a) ARI of the fitted partitions when evaluation is restricted to spots whose dominant category exceeds 0.50 or 0.75; error bars are seed SD and the spot counts are under the axis. (b) Distribution of the dominant-category proportion. (c) Gate-derived seam score against source composition entropy (seed 1); a boundary-sensitive score would rise with mixing. (d) Supplied tumor and DCIS fraction per spot.

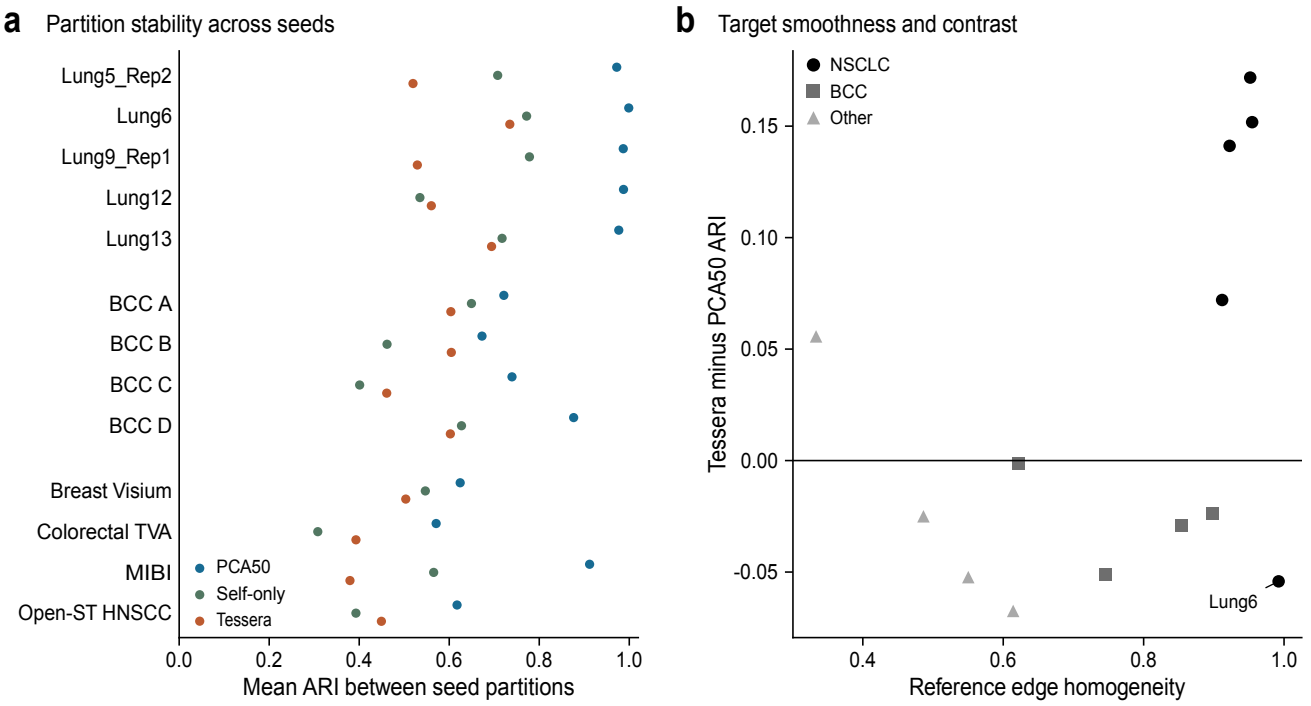


Figure S14: Optimization stability and target structure. (a) Mean ARI between the three seed partitions on identical inputs. (b) Reference edge homogeneity, the fraction of spatial edges whose endpoints share a supplied label, against the Tessera minus PCA50 contrast. Lung6 is labelled because its high homogeneity reflects a display field dominated by one region.